

PROJECT FINAL REPORT

UniShare.lk – Smart University Platform

Centralized Academic Resource Hub for Sri Lankan Universities

Web Development – Group Project

Group No. 22

Group Members

Registration Number	Index Number	Name
ITT/2024/067	2752	M.C.D.Meedeniya
ITT/2024/041	2726	H.P.H.M.Gunawardhana

27.09.2026

1. Project Overview

1.1 Introduction

UniShare.lk – Smart University Platform is a web-based academic resource-sharing system designed to provide university students with a centralized place to find, share, save and discuss academic materials. The original project proposal defines the theme as a centralized academic resource hub for Sri Lankan universities and identifies undergraduates as the primary target audience.

The completed implementation extends the proposed concept into a PHP and MySQL web application. It includes user authentication, resource uploading and downloading, resource searching and filtering, comments, bookmarks, a personal dashboard, a to-do list and a contact form.

1.2 Purpose and Goals

- Centralize academic resources in one web platform.
- Reduce the time students spend searching across multiple sources.
- Allow registered users to upload useful study materials.
- Allow students to search, filter, download and save resources.
- Support interaction through resource comments.
- Provide a personal dashboard for managing uploads, bookmarks and tasks.
- Provide a contact channel for users to communicate with the project team.

1.3 Target Audience

The main target audience is undergraduate university students in Sri Lanka. The platform is intended for students who need access to lecture notes, slides, laboratory materials, videos and past papers.

1.4 Key Features

Feature	Description
User registration	Creates a new account with server-side validation and a bcrypt-hashed password.
User login/logout	Authenticates users with sessions and protects restricted pages.
Resource upload	Allows authenticated users to upload academic files with title, category and description.
Resource browsing	Displays resources stored in the MySQL database.
Search and filtering	Searches resource titles and filters resources by category.
Download	Provides access to uploaded academic files.
Comments	Allows authenticated users to discuss resources.
Bookmarks	Allows users to save resources and prevents duplicate user-resource bookmarks.
Dashboard	Provides a personal area for uploads, bookmarks and tasks.
To-do list	Allows users to add, complete and delete personal tasks.

Contact form	Stores submitted contact messages in the database.
Responsive interface	Uses responsive CSS and Bootstrap-based layouts for different screen sizes.

1.5 Technology Stack

Technology	Purpose
HTML5	Page structure and content
CSS	Visual design and responsive styling
Bootstrap 5	Responsive UI components and layout
JavaScript	Client-side interactions and validation
PHP	Server-side processing and application logic
MySQL	Relational database management
Git	Version control
GitHub	Repository and collaboration
XAMPP	Local Apache and MySQL development environment

2. System Design

2.1 System Architecture

The system follows a client-server architecture. Users access the application through a web browser. The frontend is implemented with HTML, CSS, Bootstrap and JavaScript. PHP processes requests and communicates with the MySQL database. Uploaded files are stored in the uploads directory while their metadata is stored in MySQL.

UniShare.lk - High-Level System Architecture

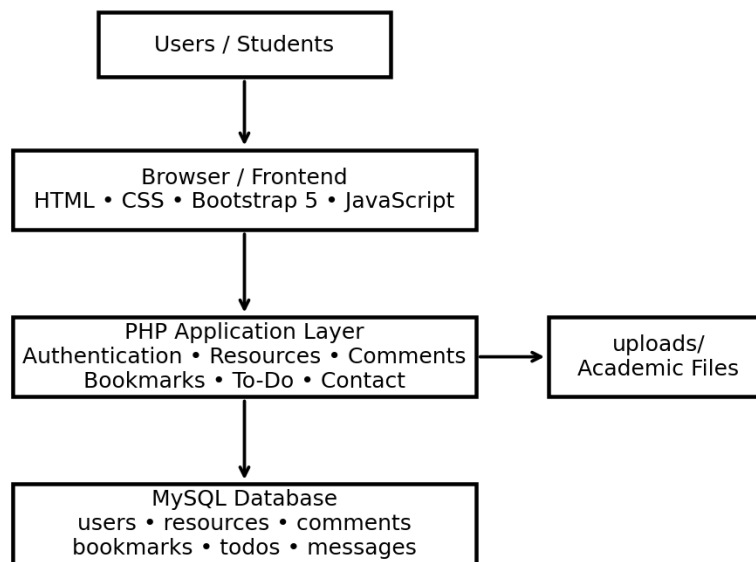


Figure 1. High-level system architecture.

2.2 Database Design

The supplied database.sql file creates the UniShare database and six main tables: users, messages, resources, comments, bookmarks and todos. Foreign keys connect user records with resources, comments, bookmarks and tasks. Resource comments reference both the resource and the user who posted the comment.

UniShare.lk – Database Relationship Overview

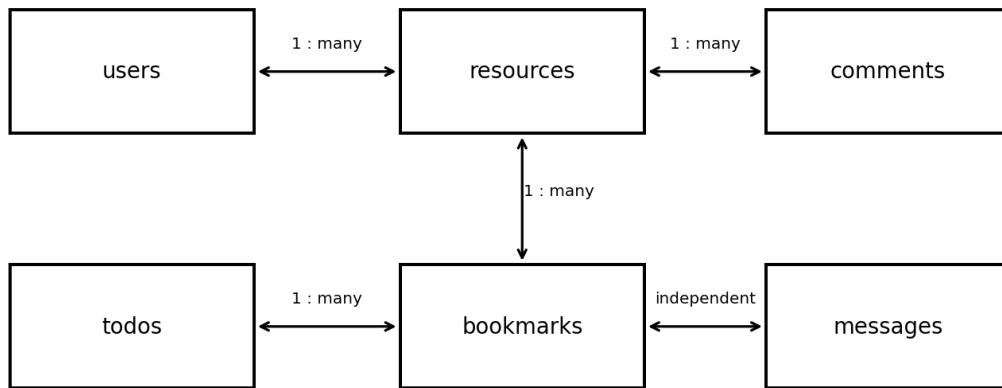


Figure 2. Database relationship overview.

2.3 Database Tables

Table	Main Fields	Purpose
users	id, username, email, password, created_at	Stores registered user accounts.
messages	id, name, email, subject, message, created_at	Stores contact-form messages.
resources	id, user_id, title, category, description, file_path, created_at	Stores uploaded academic resource metadata.
comments	id, resource_id, user_id, body, created_at	Stores comments linked to resources and users.
bookmarks	id, user_id, resource_id, created_at	Stores saved resources for users.
todos	id, user_id, task, is_done, created_at	Stores personal to-do tasks.

2.4 Database Relationships

- One user can upload many resources; each resource belongs to one user.
- One resource can have many comments; each comment belongs to one resource and one user.
- One user can create many bookmarks and one resource can be bookmarked by many users.
- A unique constraint on user_id and resource_id prevents duplicate bookmarks for the same user and resource.
- One user can create many to-do tasks.
- Contact messages are stored independently in the messages table.

2.5 Folder Structure

```
Web-Development---Group-/
├── index.php
├── dashboard.php
├── features.php
├── resources.php
├── about.php
├── contact.php
├── database.sql
├── auth/
│   ├── login.php
│   ├── register.php
│   └── logout.php
├── includes/
│   ├── db.php
│   ├── functions.php
│   ├── header.php
│   └── footer.php
├── css/
│   ├── Home.css
│   ├── Features.css
│   ├── Resource.css
│   ├── about.css
│   ├── contact.css
│   └── shared.css
├── js/
│   ├── hom.js
│   ├── resours.js
│   └── contact.js
├── images/
│   ├── logo.png
│   └── hm.png
├── uploads/
├── .gitkeep
└── .htaccess
```

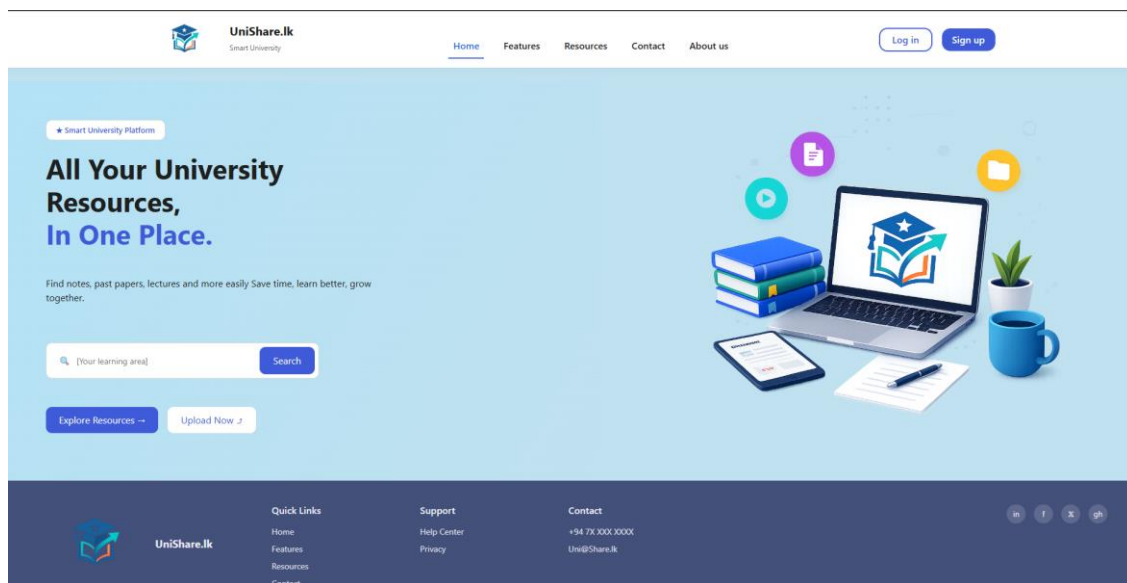
The project also contains some earlier/static HTML, CSS and JavaScript files. The final report should focus on the PHP/MySQL implementation that is used to run the completed system.

3. Implementation

This section describes the major implemented functions of the completed UniShare.lk system. Replace each screenshot placeholder with a screenshot from the running website.

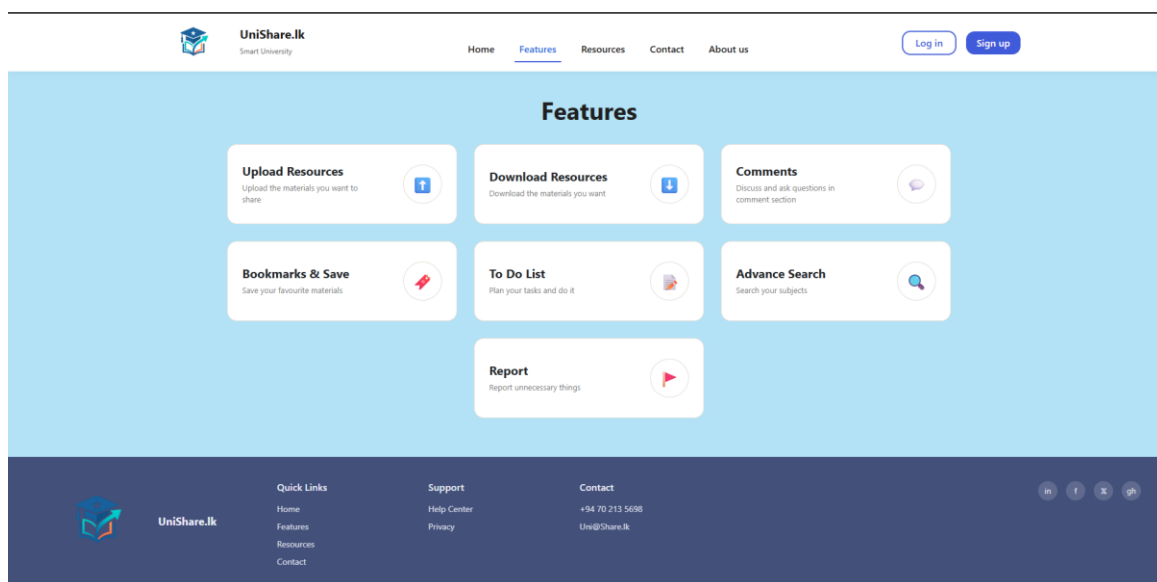
3.1 Home Page

The Home page acts as the main entry point. It introduces UniShare.lk, provides navigation to major sections and directs users toward academic resources and account functions.



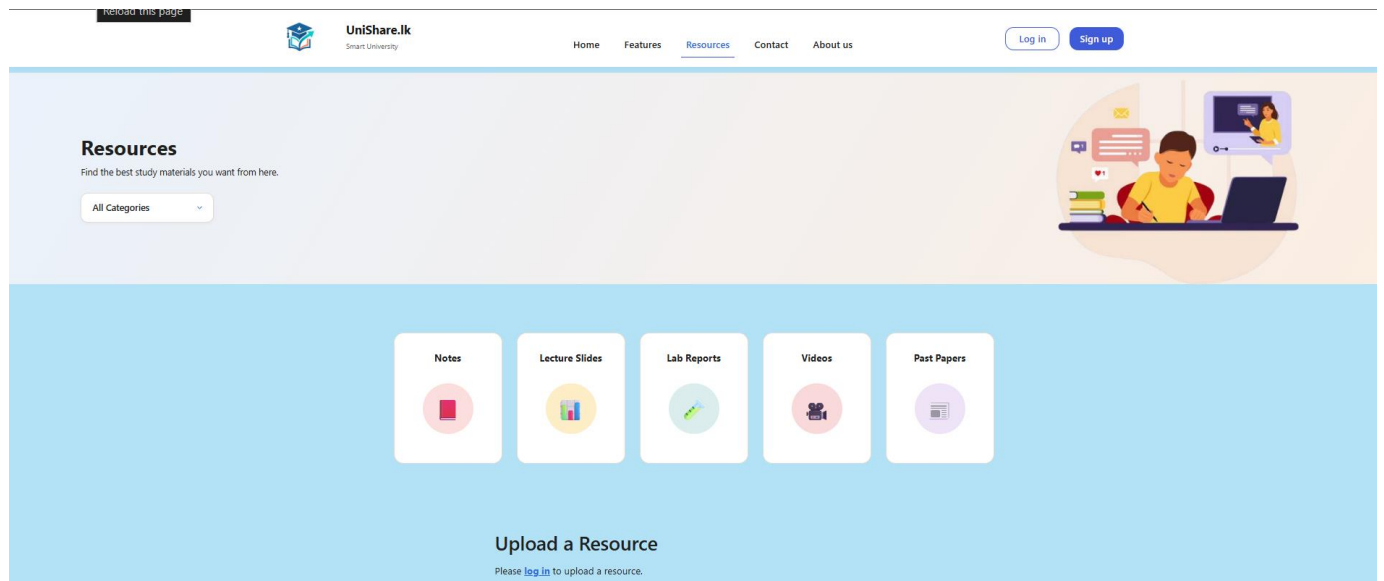
3.2 Features Page

The Features page explains the main functions provided by the platform, including resource sharing, downloading, comments and other student-focused functions.



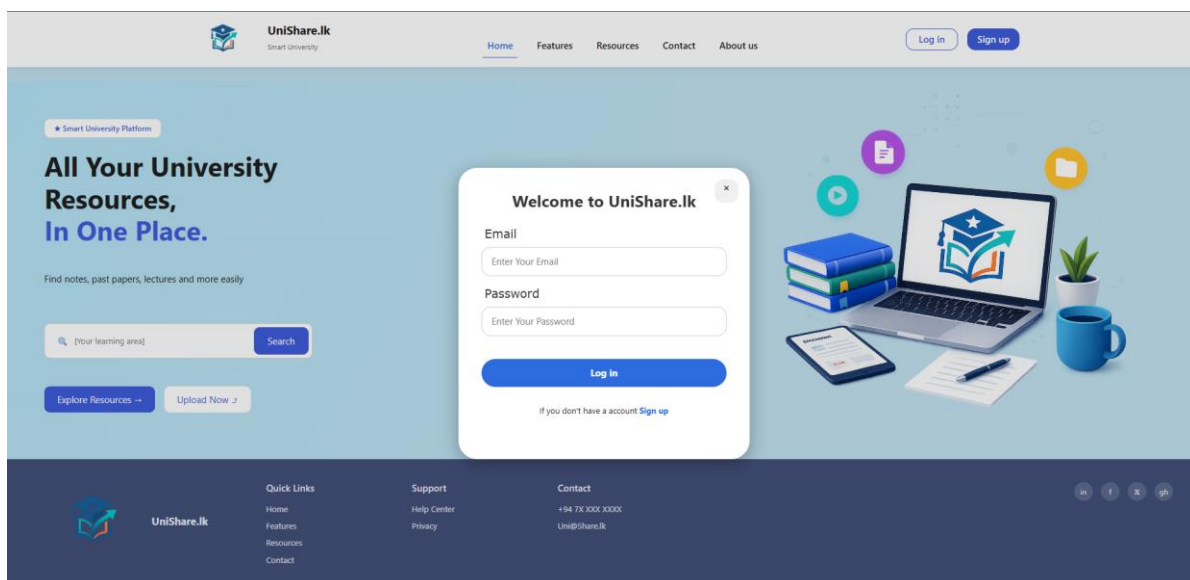
3.3 Resources Page

The Resources page retrieves resources from MySQL, displays their category, uploader and date, and provides search, category filtering, download, bookmark and comment functionality.



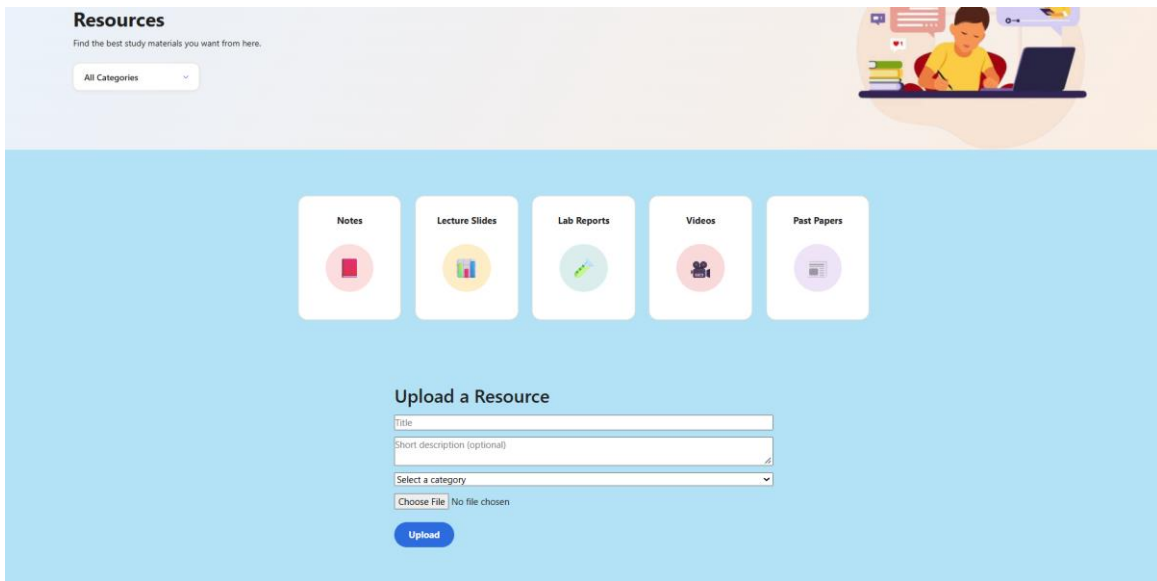
3.4 User Login and Logout

The login process authenticates a username or email with a password and establishes a PHP session. The logout process ends the authenticated session. Protected pages use a login requirement before allowing access.



3.7 Resource Upload

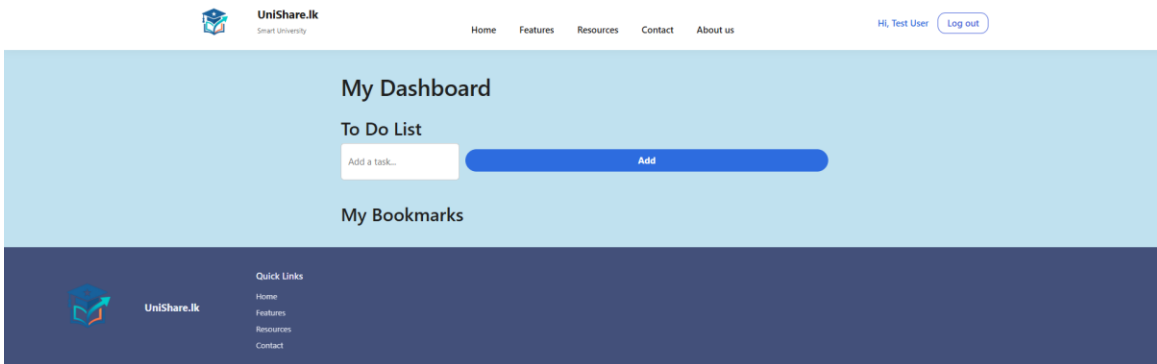
Authenticated users can provide a title, category, description and file. The uploaded file is stored in the uploads directory while the related metadata is stored in the resources table.



The screenshot shows a web interface for uploading resources. At the top, there's a header with the title "Resources" and a subtitle "Find the best study materials you want from here." Below this is a dropdown menu labeled "All Categories". To the right, there's an illustration of a student studying at a desk. The main content area has five category buttons: "Notes", "Lecture Slides", "Lab Reports", "Videos", and "Past Papers". Below these is a form titled "Upload a Resource" with fields for "Title", "Short description (optional)", and "Select a category". There's also a "Choose File" button and a "No file chosen" text. At the bottom of the form is an "Upload" button.

3.10 Bookmarks

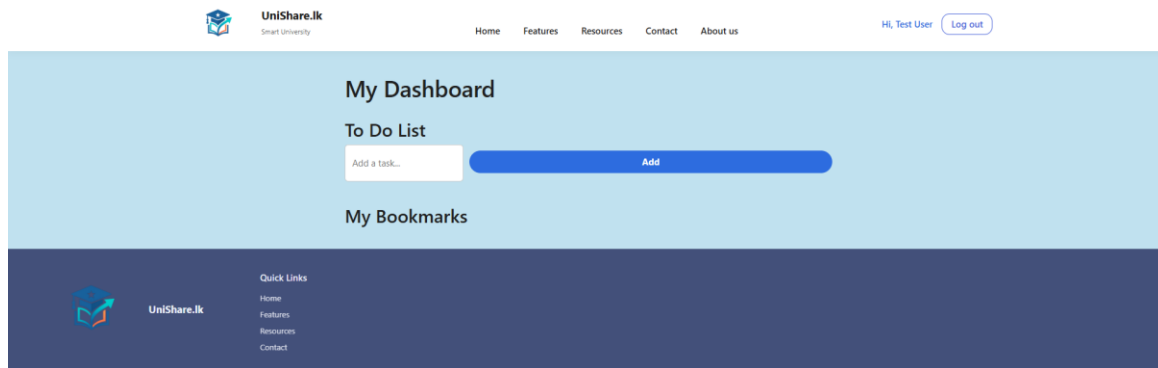
Users can save or remove resources from their bookmarks. The database uses a unique user-resource combination to prevent duplicate bookmarks.



The screenshot shows a user dashboard for "UniShare.lk". The top navigation bar includes links for "Home", "Features", "Resources", "Contact", and "About us". On the right, there's a user profile section with "Hi, Test User" and a "Log out" button. The main content area is titled "My Dashboard" and contains a "To Do List" section with an "Add a task..." input field and an "Add" button. Below this is a "My Bookmarks" section. The footer features the "UniShare.lk" logo and a "Quick Links" menu with links to "Home", "Features", "Resources", and "Contact".

3.11 To-Do List

The dashboard allows users to add tasks, toggle completion status and delete tasks. Each task belongs to the logged-in user.



3.12 Contact Form

The Contact page accepts a name, email, subject and message and stores submitted information in the messages table.

The screenshot displays the contact form on the UniShare.lk website. At the top, there are four contact information cards: 'Location' (Anuradhapura, Sri Lanka), 'Phone' (+94 70 213 5698), 'Email' (Uni@Share.lk), and 'Working Hours' (Mon - Fri, 9AM - 5PM). Below these is a 'Send us a message' section with a sub-header 'We usually reply within 24 hours.' and four input fields: 'Full Name' (placeholder: 'Your name'), 'Email Address' (placeholder: 'you@example.com'), 'Subject' (placeholder: 'What is this about?'), and 'Message' (placeholder: 'Write your message here...'). A 'Send Message --' button is at the bottom left of the form. To the right of the form is a box containing the UniShare.lk logo and the text 'UniShare.lk HQ' and '181a Campus Road, Anuradhapura, Sri Lanka'.

4. Challenges and Solutions

Challenge	Problem	Solution
Database connectivity	The PHP application requires a working MySQL connection with the correct host, database, username and password.	A centralized includes/db.php file uses PDO, exception handling, UTF-8 encoding and prepared statements.
User authentication	Restricted pages must not be accessible to users who are not logged in.	Session-based authentication and helper functions such as <code>is_logged_in()</code> and <code>require_login()</code> are used.
Password security	Plain-text passwords would create a security risk.	Passwords are stored with <code>password_hash</code> using <code>PASSWORD_BCRYPT</code> and checked with <code>password_verify</code> .
Duplicate usernames/emails	Two users should not register with the same username or email.	Server-side validation checks existing users and the database also defines unique constraints.
File uploads	Uploaded academic files must be stored while maintaining a database record for each resource.	Files are stored in <code>uploads/</code> and resource metadata, including <code>file_path</code> , is stored in MySQL.
Duplicate bookmarks	The same user should not save the same resource repeatedly.	The system checks for an existing bookmark and the database defines a unique <code>user_id/resource_id</code> combination.
SQL injection protection	User input must not be inserted directly into SQL statements.	PDO prepared statements with bound parameters are used for database operations.
Responsive interface	The website needs to remain usable across different screen sizes.	Responsive CSS and Bootstrap layouts are used to adapt page components.

5. Individual Contribution

The proposal identifies two index numbers. The project repository also contains Git commits from two contributor accounts, but the supplied materials do not provide enough evidence to map every task and index number with complete certainty. Therefore, the table below is intentionally editable. Replace the placeholders with the group's confirmed contribution record.

Member	Registration & Index Number	Contribution
M.C.D.Meedeniya	ITT/2024/067 - 2752	Resources Page Frontend and Backend, About Page Frontend and Backend, Contact Us Frontend and Backend and Bookmark & Save Feature, Reminder Feature.
H.P.H.M.Gunawardhana	ITT/2024/041 - 2726	UI/UX Design, Home page Frontend and Backend, Features page Frontend and Backend, Login Page Frontend and Backend.

6. Setup Instructions – XAMPP

1. Install XAMPP and start the Apache and MySQL modules.
2. Copy the project folder into the htdocs directory (e.g. C:/xampp/htdocs/unidock).
3. Open phpMyAdmin, create a new database, then import database.sql or unidock_exported_sql.sql to create all tables.
4. Check includes/db.php and update the database host, name, username and password to match your local MySQL setup if needed.
5. Visit {http://localhost/unidock} in a browser to run the application.
6. Register a new account and log in to access the Features, Resources, To Do List, Reminders and Q&A sections.

7. References

- Bootstrap documentation – <https://getbootstrap.com/>
- GitHub – <https://github.com/>
- Font Awesome 6.7.2 – <https://fontawesome.com>
- I used it to further my learning of HTML, JS, and PHP. – <https://www.w3schools.com>